

Price Prediction for Flights & 5* Hotels from NY Airports to Cancun

Gavin Garcia (geg80), Alvin Jeon (atj5), Bismah Ahmed (ba562)

- 1. Project Description:** A machine learning predictive model to help maximize one's budget for a vacation in Cancun, utilizing trends to inform users on the best booking time for the best price and accommodations.

1.1. Problem Statement

The main issue that the creation of this project hopes to solve is the prediction of price fluctuations in real-time for booking flights to Cancun alongside a 5* hotel resort. Cancun is a world-famous travel destination, with countless people booking reservations months or years ahead of time, and the issue becomes timing the booking correctly based on a wide variety of factors, including but not limited to: plane prices, weather conditions, tourist density, crime rate, hotel promotions, and hotel/excursion accessibility. We aim to create a model that aggregates information together from the various factors that affect booking prices to produce both a compelling product and project, as well as build a useful tool to incorporate into future planning when traveling with friends and family.

1.2. Connection to Course Material

This project covers multiple topics from our course, including Python data structures, Numpy, Pandas, DataFrames, data collection and transformation through the Requests, csv, and Pandas libraries, visualization through Matplotlib and Seaborn, and predictive modeling through machine learning in Scikit. We used Pandas to import the Cancun weather dataset, clean the data, rename columns, remove missing values, and organize the data into a dataframe. We used Numpy for most of the numerical work, for example, adding randomness to our synthetic flight and hotel prices to make them look more realistic and generating correlation matrices for weather indicators with the `.corr()` command. For predictive modelling, we used the Random Forest Regressor in the Scikit library to predict hotel and flight prices based on factors such as the archetypically cyclical weather patterns and other seasonal and holiday trends. Overall, we covered most of the major code libraries that were focused on in class, allowing us to build a multifaceted project using resources within our scope for this class.

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2. Novelty and Importance:

2.1. *Significance of the Project*

International travel bookings are planned months in advance. With that in mind, how can travelers on a budget know the best time to book and what the weather conditions are during those months without extensive research? As a group of travelers who have spent hours creating one dubiously cost-effective itinerary after another, only for the price to drop further at some other time before or after we booked, the idea of a predictive model to better inform our decision-making was the perfect candidate. Combining personal experience and market trends with machine learning, a modeling system that indicates the timing of trips while accounting for conditions that can make or break an experience will offset the hours of research that go into traditional travel planning. Savvy users can derive insights that can save them hundreds of dollars and double their money's worth in relief from the stresses of travel planning. Our predictive model will give users much more clarity on the price trends of flights and hotels in Cancun into the future, using recently scraped data from 2023-2026 to analyze trends in weather and how they correlate to price changes in the two most costly aspects of a high-end Cancun vacation.

2.2. *Relevance*

Traditional airline websites and hotel booking platforms mostly list current prices but fail to offer insights for when prices are likely to increase or decrease. They aren't incentivized to, as they would rather consumers buy travel packages when they are more expensive, for the sake of their own profit. Consequently, travelers often have to search manually and guess the best time to book their flights. By endeavoring to create a prediction model in this field, we are mechanizing the highly speculative world of travel pricing for a tool meant to serve consumers. Moreover, the hotel prices and flight prices can fluctuate due to seemingly unpredictable external factors such as hotel promotions, world events, and weather conditions. These fluctuations can be hard to detect, affect travel prices, and introduce noise to the prediction model while also affecting prediction accuracy. Our predictive model stays relevant even with these factors, as we factor in

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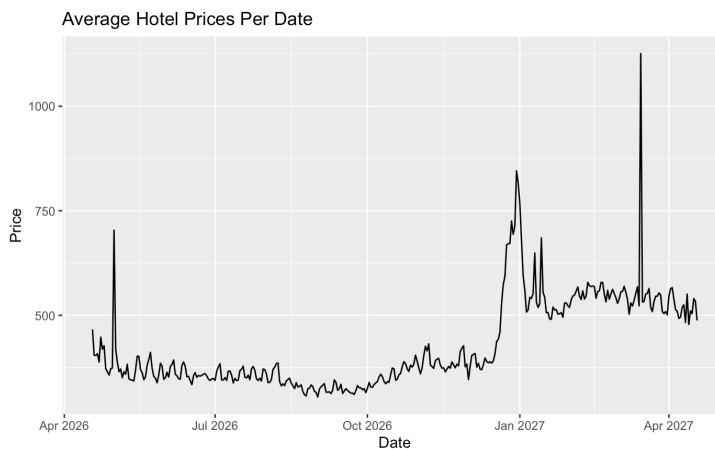
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random noise from unexpected shocks alongside holiday trends and weather patterns that also greatly vary flight and hotel prices.

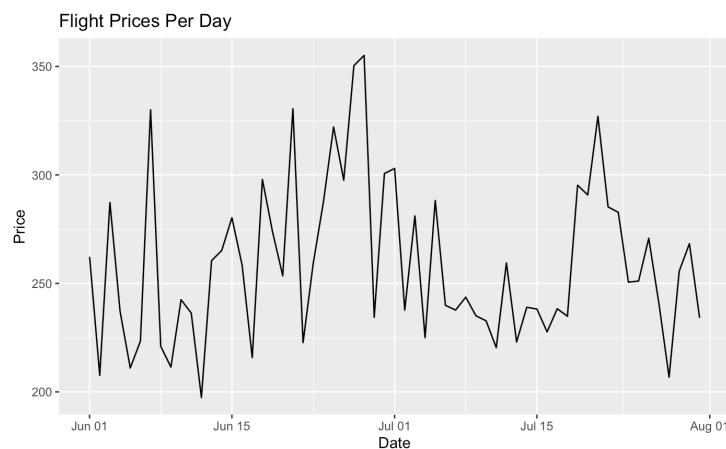
3. Progress and Contribution:

3.1. Data Utilization

The hotel, flight, and weather datasets consisted of real price data collected from different sources, which were initially used to visualise and analyze trends.



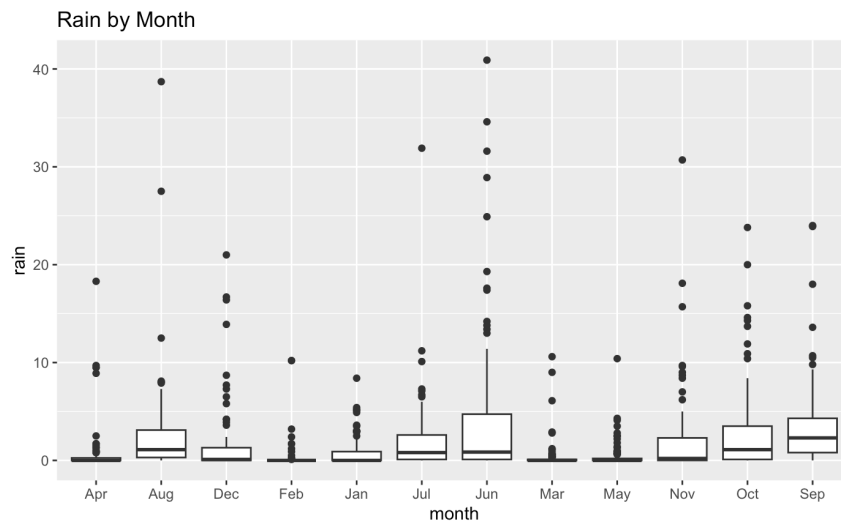
We can clearly see the price varies across different dates, which means that season and timing affect hotel prices. This visualization was generated from our `cancun_hotel_prices.csv` file in our `Visualization.rmd` script. It demonstrates to us a general curve for hotel prices from our collected data of April 2026 to April 2027, which we pasted backwards



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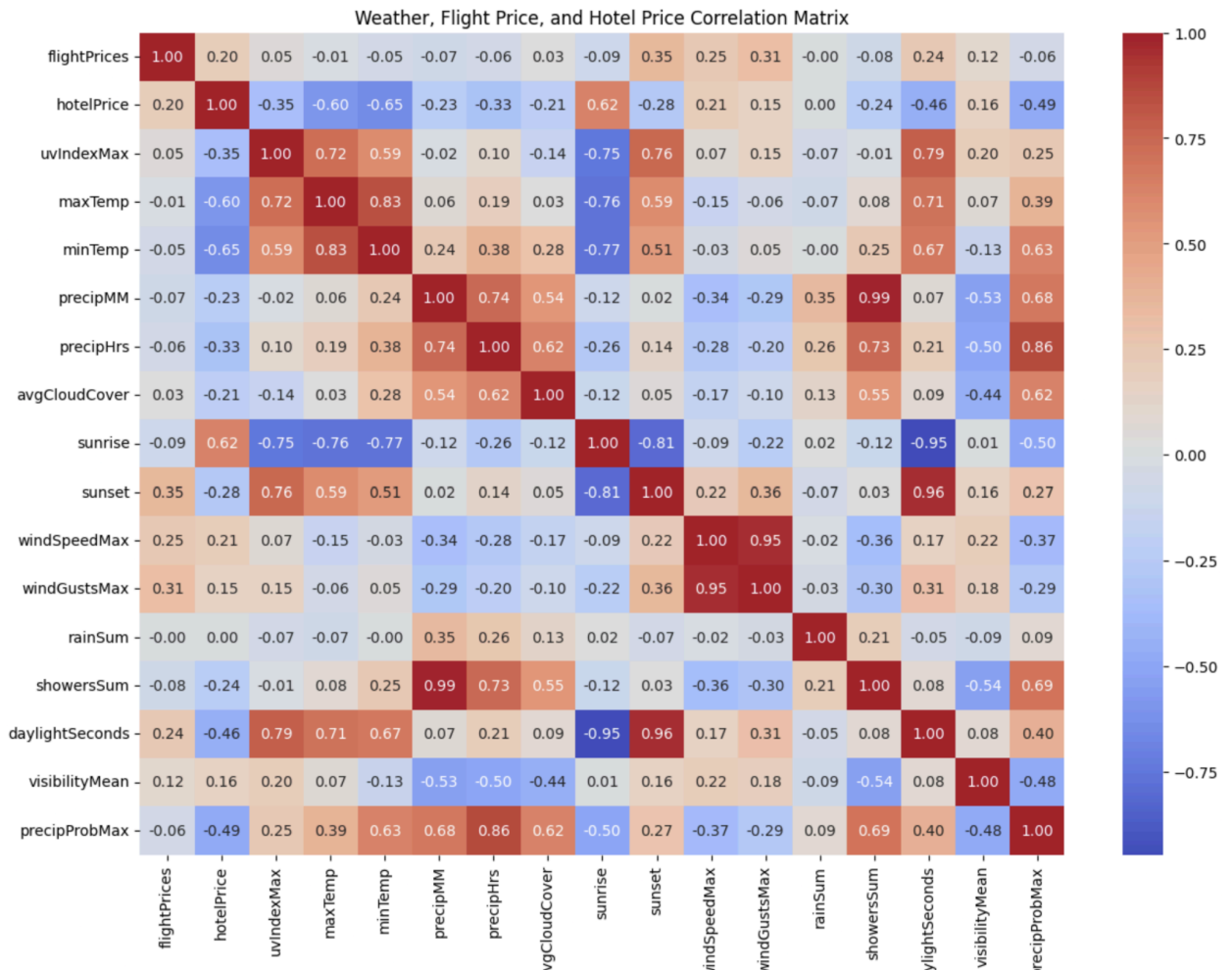
This graph shows only 2 month flight prices for 2026, and mostly during these months, flight prices tend to be higher. However, the graph shows frequent fluctuations, which means that prices can change a lot in a short time. This data can be used to create more realistic synthetic data for more accurate predictions. We copied back these two months of data from 2026 to our 2023-2025 range (the year range of the dataset we would be using to train our machine learning model), and used interpolation and bfill/ffill to create a sloping flight price curve that we would then influence with correlation-matrix-derived weather multipliers, seasonal modifiers, and random noise.



This boxplot shows that from December to May, there is less rainfall, while from June to November, there is more rain. The extreme points can be seen in June and August, which suggests that this might be the precipitation season, so it might impact travel planning and prices, which could be due to flight cancellations, etc. Rain was one of the factors we believed could potentially influence flight and hotel prices, so we built a correlation matrix of many different weather factors we found in our `cancun_weather_final.xlsx` spreadsheet pulled from an open-source API, as historical weather data isn't really proprietary information, unlike past hotel and flight prices, which we were only able to gather limited amounts of data. Some of these weather factors include UV Index, low and high temperatures, precipitation levels, amount of time, sunrise and sunset times (total daylight), cloud cover (average visibility), and precipitation probability.

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Below was the correlation matrix that we generated by running the .corr command on our merged dataset across all weather columns (defined as correlationColumns). By observing the top two rows indicated as “flightPrices” and “hotel Price”, one can notice that the more saturated blues and oranges indicate a stronger correlation between a certain weather factor and the price of a flight or hotel. We averaged the correlation values of each weather factor across hotel and flight prices and compared it against a correlation threshold of 0.2. If the average of a weather factor was above this correlation threshold, the data values we scraped for that weather factor were included as input columns for our machine learning model. Otherwise, if the weather factor did not have a strong enough average correlation, it was dropped.

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3.2. *Models, Techniques, Algorithms*

The model that we used for our machine learning dataset was the Random Forest Regressor from Scikit-learn in order to generate predicted hotel and flight prices for 2027.

The technique we used to gather our data was utilizing the Requests library to help us make API calls in both the flight price .py script and the hotel prices .ipynb script. We used the CSV library to convert our scraped data into .csv files, which were easily viewable in Codebench and easily imported into DataFrames for analysis and manipulation. This is how we collected a majority of our more difficult data. The specific functioning of the algorithm in the RandomForestRegressor is essentially a number of decision trees trained on a random portion of the data and aggregated together as part of a bootstrap aggregating technique.

We used the Pandas and Numpy libraries in all of our scripts to help capture and clean our scraped data from various formats (either .xlsx or .csv), with .interpolate helping us get a starting curve for our flight price data, which we would then vary with the correlation matrix for weather factors that we built and plotted with .corr. Matplotlib and Seaborn served as our primary visualization techniques to help us see our data every step of the way, providing us with an easy-to-understand correlation matrix, line graphs, and other types of plots, helping us view our multi-faceted and complex scraped, cleaned, and synthesized data from multiple different angles.

3.3. *Experimental Design*

We quickly discovered at the beginning of our project that historical hotel and flight price data were heavily guarded and highly proprietary data secrets of the we used these datasets that we were able to find and turn them into accurate synthetic data through correlation matrices, random noise, seasonal multipliers, and some interpolation.

Photo of Hotel CSV File:

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	date	hotel_name	hotel_key	ota	price_usd
1	2026-04-18	GR Solaris Cancun	g150807-d501418	Booking.com	179
2	2026-04-18	GR Solaris Cancun	g150807-d501418	Trip.com	220
3	2026-04-18	GR Solaris Cancun	g150807-d501418	Agoda.com	171
4	2026-04-18	GR Solaris Cancun	g150807-d501418	Vio.com	166
5	2026-04-18	GR Solaris Cancun	g150807-d501418		
6	2026-04-18	GR Solaris Cancun	g150807-d501418		
7	2026-04-18	GR Solaris Cancun	g150807-d501418		
8	2026-04-18	GR Solaris Cancun	g150807-d501418		
9	2026-04-18	GR Solaris Cancun	g150807-d501418		
10	2026-04-18	GR Solaris Cancun	g150807-d501418		
11	2026-04-18	GR Solaris Cancun	g150807-d501418		
12	2026-04-18	GR Solaris Cancun	g150807-d501418		
13	2026-04-18	GR Solaris Cancun	g150807-d501418		
14	2026-04-18	GR Solaris Cancun	g150807-d501418		
15	2026-04-18	GR Solaris Cancun	g150807-d501418		
16	2026-04-18	GR Solaris Cancun	g150807-d501418		
17	2026-04-18	GR Solaris Cancun	g150807-d501418		
18	2026-04-18	GR Solaris Cancun	g150807-d501418		
19	2026-04-18	GR Solaris Cancun	g150807-d501418		
20	2026-04-18	GR Solaris Cancun	g150807-d501418		
21	2026-04-18	GR Solaris Cancun	g150807-d501418		
22	2026-04-18	GR Solaris Cancun	g150807-d501418		
23	2026-04-18	GR Solaris Cancun	g150807-d501418		
24	2026-04-18	GR Solaris Cancun	g150807-d501418		
25	2026-04-18	GR Solaris Cancun	g150807-d501418		
26	2026-04-18	GR Solaris Cancun	g150807-d501418		
27	2026-04-18	GR Solaris Cancun	g150807-d501418		
28	2026-04-18	GR Solaris Cancun	g150807-d501418		
29	2026-04-18	GR Solaris Cancun	g150807-d501418		
30	2026-04-18	GR Solaris Cancun	g150807-d501418		

For scraping hotel price data, we used a script assembled through [Xotelo.com](https://www.xotelo.com/) (which had resources and script snippets for scraping such data) that helped us develop a `cancun_hotel_prices.ipynb` script for scraping daily prices across about 35 different 5-star-classified hotels from 4 online travel agencies + the hotel's own website. We were able to scrape data in this manner from 4/18/2026 to 4/18/2027, with every single day in that dataset containing up to 175 price entries from different online travel agencies and 5-star hotels. The reason why we had to collect the data for each hotel from multiple online travel agencies is that each hotel has different prices on each website, easily observable in the first few lines of the CSV, where GR Solaris Cancun has prices ranging from 166 to 220 USD across four different online travel agencies. This is where services like Trivago make their money, as their whole service proposition is helping consumers compare differing prices for the same hotel across multiple booking websites. We essentially used our data scraping script to perform a similar

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function of aggregating hotel prices, except we channeled these prices into an average daily prices for 5-star hotels in Cancun.

Photo of Flight Price Data CSV file:

	date	price
1	6/1/26	159
2	6/1/26	164
3	6/1/26	178
4	6/1/26	253
5	6/1/26	259
6	6/1/26	292
7	6/1/26	531
8	6/2/26	159
9	6/2/26	164
10	6/2/26	195
11	6/2/26	220
12	6/2/26	300
13	6/3/26	159
14	6/3/26	164
15	6/3/26	169
16	6/3/26	253
17	6/3/26	272
18	6/3/26	333
19	6/3/26	661
20	6/4/26	190
21	6/4/26	230
22	6/4/26	271
23	6/4/26	293
24	6/4/26	164
25	6/4/26	195
26	6/4/26	228
27	6/4/26	235
28	6/4/26	328
29	6/5/26	204
30	6/5/26	215
31	6/5/26	164
32	6/5/26	174
33	6/5/26	178
34	6/5/26	190
35	6/5/26	235
36	6/5/26	328

We aggregated our flight data much more simply, including just a date and various price points that our limited API scraper (DataFinalFlightCode .py script) was able to find. We averaged these price points together to get a mean price for each day, which we then pasted across all years of our training data and built a curve around for the rest of the year's flight price data. Finally, we used Kaggle's interactive weather data API to gather comprehensive weather data for Cancun across our entire date range.

From there, our primary experimental design for analyzing the data included using Pandas, Scikit, Matplotlib, and Seaborn, to generate a correlation matrix for weather factors to flight and hotel prices, train a machine learning model on strongly correlated data (determined by our correlation matrix), and plot various aspects of the scraped and ML-generated for visual analysis.

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3.4. Key Findings and Results

The largest key findings here included the weather factors that most heavily influenced flight and hotel prices. Originally, we just assumed the amount of impact that certain weather factors had on prices and built functions to artificially program this bias in, when, in fact, this analysis of our own wasn't actually supported by the real correlation scores that we generated.

	date	flightPrices	hotelPrice	maxTemp	minTemp	sunrise	sunset	windSpeedMax	windGustsMax	daylightSeconds	precipProbMax
0	2023-04-18	283.916516	488.754065	89.1	69.2	23100	68820	7.4	21.0	45703.93	0.0
1	2023-04-18	283.916516	510.948539	89.1	69.2	23100	68820	7.4	21.0	45703.93	0.0
2	2023-04-19	286.506204	436.155018	84.2	74.8	23040	68880	11.6	29.3	45787.48	NaN
3	2023-04-20	268.564913	432.945136	87.5	77.4	23040	68880	12.3	29.5	45854.17	NaN
4	2023-04-21	272.636548	423.138246	86.8	77.0	22980	68880	10.4	25.7	45920.64	NaN
...	...	...	...	...	...	...	...	...	...	...	...
95	2023-07-21	322.453803	301.614247	91.1	76.8	22620	70200	7.1	20.8	47608.53	65.0
96	2023-07-22	296.529916	305.477765	92.3	76.7	22620	70200	7.6	19.9	47565.46	73.0
97	2023-07-23	251.518358	284.210127	89.6	75.8	22680	70200	6.2	15.9	47521.49	98.0
98	2023-07-24	216.063337	287.778178	91.1	76.6	22680	70140	6.5	17.7	47476.67	78.0
99	2023-07-25	250.978215	333.545155	90.4	76.5	22680	70140	9.4	24.4	47431.05	43.0

We had predicted that the temperature, amount of rain and level of cloud cover would be the largest factors in varying flight and hotel prices, when only temperature was found as a strongly correlated factor to the prices. Instead of the amount of rain correlating the prices, the probability of precipitation for that day had a stronger effect on prices. Additionally, the time of sunset and average/gust wind speed had similarly strong correlations (either positive or negative) to both flight and hotel prices, so they passed the threshold that way. Meanwhile, factors such as min and max daily temperatures, amount of daylight, and precipitation probability were very strongly correlated to hotel prices (either positively or negatively), which passed them over our correlation threshold.

3.5. Evaluation

To evaluate our results, we can now say more confidently that the factors most correlated to varying the prices of Cancun hotels and flights are the amount of daylight, precipitation probability, wind speed, and temperature. With this in mind, economically minded families who want luxury for less can book on dates

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with higher predicted precipitation probability, on days where it's predicted to be a bit less windy (more dry heat) and cooler. Given Cancun's typically warmer weather, these days would still be relatively mild and enjoyable, and prevent some of the biggest positively correlated factors to hotel price jacking up the price of their stay. Finally, if you want to factor in seasonality as well, September and October are attractive travel times for lower prices, as it is sandwiched between the high seasons of the summer months and December. If you're willing to bet on the forecast for rain to be wrong, that may also play in your factor in making an economical decision. All in all, the weather factors that affected hotel and flight prices in Cancun are a little more nuanced than immediately meets the eye.

3.6. *Advantages and Limitations*

The advantage of using our predictive price model over other price models is that it is built from real price data that is actually correlated to hotel and flight prices, effectively taking the guesswork out of determining what sort of weather conditions could help you save a couple dozen, if a few hundred dollars on hotels over the course of your Cancun stay. Additionally, our solution also factors in seasonality and specific occasions to avoid (should consumers wish to avoid price spikes) such as Spring Break in March and the holidays from late December into early January. Additionally, our price predictive model produces a nice little text summary of the predicted costs of your stay for you, factoring in day-to-day variations in flight prices as it grabs data at the beginning and end of your stay and aggregates the prices together in an easy-to-display format for solo travelers and couples!

A summary of a trip's cost during low season at the beginning of September:

```
Trip Summary
Arrival: September 01, 2027
Departure: September 06, 2027
Nights: 5

Flight Prices
• One-way (1 person): $164.99
• Round-trip (1 person): $329.47
• One-way (2 people): $329.99
• Round-trip (2 people): $658.94

Hotel Prices
• Total hotel cost: $1276.94
• Price per night: $255.39
```

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However, our model does have some important limitations in its findings, as its training data only comes from recent price trends from 2023 through 2026, which means that our gathered data is only likely relevant for predicting prices in 2027-2029, before inflation and other longer-term economic, environmental, or cultural factors start fundamentally altering the way that people travel to Cancun and other destinations around the world. Additionally, due to the heavily proprietary nature of historical flight and hotel price data, we needed to use limited and freeware API scraping tools to find a limited amount of data that was actually from 2026-2027, which we had to use to inform synthetic price data from 2023-2026 that would be then correlated to the complete weather data that we were to find, meaning that there is mostly a simulation of historical prices that our predictive price model trains on. This could affect the overall accuracy of our model. Finally, the last major limitation of our model is that it only covers the prices of flights from Newark to Cancun, which means that it specifically serves Northeastern and Tri-State travelers who want to fly to Cancun, and not many other populations looking for travel.

4. Changes After Proposal:

4.1. *Differences from Proposal*

The main difference that derived from the Proposal was the belief in idealistic websites for data gathering and analysis. Initially, the collection of data seemed possible from open-source websites to determine current prices of hotels, flights, and weather distribution. However, after looking into the provided information of the sites themselves, many kept them behind locked paywalls. This caused a shift in focus in how data would be accumulated. For weather data, past historical API data had to be utilized. For hotels, a third-party API data-gathering source was needed, where a member would have to manually enter all the keys and create a CSV data file from scratch. Flight prices also relied on non-traditional sources, gathering data from outside sources that delved into actual flight patterns. This ultimately meant that we were not able to scrape and gather data for our planned data range of April 2023 to April 2026, which we were going to train our machine learning model with in order

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4.2. Challenges

One major difficulty we faced was collecting reliable flight prices. First, we tried using the Bureau of Transportation Statistics because it provides official transportation datasets, but the datasets were extremely large and not directly useful, as they mainly focused on domestic U.S. flights. Due to this, we used Google Flights API to extract future flight prices, giving us a more realistic idea of price trends. However, the API had usage limits, so we were only able to collect flight prices for June and July 2026, which were then used as a reference to create synthetic flight prices.

5. Conclusion and Future Work:

5.1. Summary of Contribution

Bismah was responsible for the collection of flight data and a great deal of the final report. This was an important role as the flight data is a difficult piece of data to collect due to limited access and large datasets. She experimented with multiple websites and data sources, but faced challenges to find relevant data so to overcome this she wrote API based code to gather real flight prices. After that she used real datasets to visualise and analyse trends initially. Furthermore, due to her well-versed analytical and writing skills, she was able to fill a lot of the reports. Finally, Bismah helped explain the script she used to scrape flight prices for the data collection portion of the project described in the demo video.

Gavin was responsible for the main code that gathered information from the flight, hotel, and weather reports to develop a machine learning algorithm that would predict prices considering all the factors together. With stronger coding ability, he decided to take on the responsibility alongside insight from other members to work on the main piece of the project. Additionally, Gavin also facilitated and recorded a majority of the demo video, including editing the video upon its completion. Furthermore, Gavin also reached out to TA's for feedback during the TA project help session and organized the files in the final project

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submission. Finally, Gavin edited the final project report and completed most of Question 3 with the help of Bismah's code visualisations.

Alvin was responsible for the discovery and application of weather historical data and hotel prices. With hotels also being locked behind private data, he had to search through and manually input all the hotel keys, developing a coding program to iterate through each to produce a manageable CSV file. Furthermore, with a program that would take multiple hours, he took on the role to run it behind the scenes for the development of a precise CSV file. Alvin also assisted with the demo video and explained his API calling script for hotel data prices.

5.2. *Future Direction*

For future improvements, we can incorporate more real-world data instead of relying on synthetic data. Furthermore, if continued, the use of actual data sets can be purchased and utilized within the code to determine prices more accurately. Currently, a lot of machine learning and estimation is being derived as the price fluctuates wildly for hotel prices, weather, and flight prices. As such, if those averages are more concise within the data sets, the analysis by said machine is able to be fully optimized and imported into realistic production. Finally, we could also expand our text results for the end user to be more interactive, including a selectable calendar of dates, which airport they wish to depart from, and potentially a desired destination from a list of destinations beyond solely Cancun. That way, the Cancun 5-Star-Stay Price Predictive Model can reach beyond its original bounds and serve more users.